

(SA1) Lego Designers: Research		MYP Design Grade 6
Statement of Inquiry:	Designers must develop research, planning, and evaluation skills to create good products.	

Students will complete research tasks to develop their new Lego product idea. This will include: 学生将完成研究任务以开发其新乐高产品创意。具体包括：

1. A **research plan** which explores Lego products and if a new set would be a good idea. 一份研究计划，探讨乐高产品并评估推出新套装的可行性。
2. A **statement** explaining why a new set is needed. 一份说明，阐述推出新套装的必要性。
3. An **exploration** of an **existing** Lego set that inspires your idea. 对现有乐高套装的分析，阐明其如何激发你的创意
4. An **outline** of what kind of set you will plan. 拟定套装的规划纲要。

You will use the Lego website and Brickipedia for your research. Links are in Moodle.

Work to be completed: 待完成工作：	Finish the tasks in this paper, including the research plan , your statement , the product exploration , and an outline of your idea. 完成本文中的任务，包括研究计划、个人陈述、产品探索以及创意大纲。
Number of lessons for work: 工作所需课时数：	4
Where to submit the work: 作品提交地点：	Turn in your paper in class. Take a picture and upload to ManageBac. 在课堂上提交你的论文。拍照上传至ManageBac系统。
Special Requirements: 特殊要求：	No AI, no use of translators for anything more than 1 word. (Cannot translate sentences or more) 禁止使用AI，禁止使用翻译工具处理超过1个单词的内容。（不得翻译句子或更长文本）

During this task students will develop the ATL skill of **research** by exploring Lego products, justifying the need for a solution, and writing research questions. 在此任务中，学生将通过探索乐高产品、论证解决方案的必要性并撰写研究问题，培养研究型学习技能（ATL）。

All your writing must be completed in English. (use of *some* pinyin allowed if absolutely necessary, no hanzi) 所有写作必须使用英语完成。（如绝对必要，允许使用少量拼音，不得使用汉字）

Task Specific Clarifications 任务特定说明

Part 1: Aii. State and prioritize the main points of research needed to develop a solution to the problem
Write 3 research questions to help you learn about Lego sets and choose the right idea for a new set. Find the answers to your research questions. 阐述并优先排序解决该问题所需的主要研究要点撰写3个研究问题，以帮助你了解乐高套装并选择新套装的合适创意。找出这些研究问题的答案。

Part 2: Ai. Explain and justify the need for a solution to a problem
Explain and justify the why it is a good idea to design a new Lego set. 阐述并论证解决该问题的必要性阐述并论证设计新乐高套装的合理性。

Part 3: Aiii. Describe the main features of one existing product that inspires a solution to the problem
Describe the features of a Lego set that you could use as inspiration for your design. 描述现有产品的核心特征，阐明其如何启发解决方案描述某款乐高套装的特征，说明其如何成为你设计的灵感来源

Part 4: Aiv. Present the main findings of relevant research.
Share the important details of the idea you are planning to develop. 呈现相关研究的主要发现

Part 1: Write **3 research questions** about Lego products to help you learn what already exists, and to help you figure out what you want to design. After you write the questions you must use **Lego.com** and **Brickipedia** to do your research and find answers. Write what you learned after each question. 围绕乐高产品撰写3个研究问题，以帮助你了解现有产品，并明确你想要设计的内容。完成问题撰写后，必须使用Lego.com和Brickipedia进行研究并寻找答案。每个问题后请记录你的研究成果。

Example Research Questions: Are there any Lego sets about basketball? 有没有关于篮球的乐高套装？ What kinds of sets are included in the Lego Icons theme? 乐高Icons系列包含哪些套装？ What themes have Lego made that are related to science fiction? 乐高公司推出过哪些与科幻相关的主题系列？ Have Lego ever made Percy Jackson sets? 乐高公司是否曾推出过珀西·杰克逊主题套装？

Question 1 问题1: _____

What I learned 我学到了： _____

Question 2 问题2： _____

What I learned 我学到了： _____

Question 3 问题3: _____

What I learned 我学到了： _____

Part 2: Write a **statement** explaining why a new set is needed. Follow the guide in Moodle to help write your statement, and be **detailed!** 撰写一份说明，阐述为何需要新增课程。请参照Moodle平台中的指南撰写说明，内容需详尽！

Examples: *a new Lego set is needed because...* 示例：需要一套新的乐高积木套装，因为.....

- The theme doesn't have anything like it... 这个主题没有类似的功能.....
- There are no other themes like my ideas... 没有其他主题能实现我的想法.....
- There are no sets available in a theme for a specific price... 主题中没有按特定价格提供的套装.....

Explain why it is a good idea to create a new set 解释为什么创建一个新集合是个好主意：

Part 3: Investigate Lego products to learn about how Lego designers create sets, and to find ideas which will help you to make your own set as good as possible. 研究乐高产品，了解乐高设计师如何设计套装，并从中获取灵感，帮助你尽可能完善自己的套装设计。

Complete the tables for 4 different Lego sets. 为4款不同的乐高套装填写表格。

EXAMPLE SET -示例集					
Set Name 名称集:	The Lord of the Rings: The Shire				
Item Number项目编号:	10354	Pieces件数:	2017	Theme主题:	Lego Icons
Features功能:	• There are many small details from the Lord of the Rings movie included, like Gandalf's Cart and a Dragon Rocket. • Created for older builders and adults. • The inside of the Hobbit house is very detailed, so people can use their imagination when making scenes.				
My observations我的观察:	• This set is part of the 'Lego Icons' theme, even though they make a Lord of the Rings theme. I think it's because it is very big and expensive. • It says '18+' for the age, but I think that younger people can also build it. Maybe they mean 18+ because it is expensive and they think adults would like it more. • Pieces are put together in a lot of really creative ways.				

Ideas I could use for my design我设计中可以借鉴的创意:	- I like how they used slope pieces for the side of the hill, and I think I'll use them for my design. - I want to have some secret parts of my design too. - I think the tree looks very unique and realistic, and I want to try making trees using the pieces they used. - The banner that says Bilbo is 111 is cool, and I want to learn to use Part Designer to make custom pieces like that for my set.
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Lego Set 1 -乐高套装1					
Set Name 名称集:					
Item Number项目编号:		Pieces件数:		Theme主题:	
Features功能:					
My observations我的观察:					
Ideas I could use for my design我设计中可以借鉴的创意:					

Lego Set 2 -乐高套装2					
Set Name 名称集:					
Item Number项目编号:		Pieces件数:		Theme主题:	
Features功能:					
My observations我的观察:					

Ideas I could use for my design我设计中可以借鉴的创意:	

Lego Set 3 -乐高套装3					
Set Name 名称集:					
Item Number项目编号:		Pieces件数:		Theme主题:	
Features功能:					
My observations我的观察:					
Ideas I could use for my design我设计中可以借鉴的创意:					

Lego Set 4 -乐高套装4					
Set Name 名称集:					
Item Number项目编号:		Pieces件数:		Theme主题:	
Features功能:					

My observations我的观察:	
Ideas I could use for my design我设计中可以借鉴的创意:	

Part 4: Explain the ‘**big idea**’ for what you want to create, and **summarize** your research and how it will help you design your set 阐述你想要创作核心理念，并总结你的研究成果及其如何助力布景设计:

The **theme** I am planning to choose is 我计划选择的主题是 _____

I chose this theme because 我选择这个主题的原因 _____

My **specific idea** is to create a set about 我的具体构想是创建一个关于.....的系列。 _____

When I researched Lego sets I found 在研究乐高套装时，我发现 _____

It is a **good idea** to design this set because 设计这套方案是个好主意，因为 _____

IB Criteria

1–2

The student:

- i. states the need for a solution to a problem
- ii. states the findings of research.

3–4

The student:

- i. outlines the need for a solution to a problem
- ii. states some points of research needed to develop a solution, with some guidance
- iii. states the main features of an existing product that inspires a solution to the problem
- iv. outlines some of the main findings of research.

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The student:

- i. explains the need for a solution to a problem

- ii. states and prioritizes the main points of research needed to develop a solution to the problem, with some guidance
- iii. outlines the main features of an existing product that inspires a solution to the problem
- iv. outlines the main findings of relevant research.

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The student:

- i. explains and justifies the need for a solution to a problem
- ii. states and prioritizes the main points of research needed to develop a solution to the problem, with minimal guidance
- iii. describes the main features of an existing product that inspires a solution to the problem
- iv. presents the main findings of relevant research.

DESIGN COMMAND TERMS

Command term	Definition
Analyse	Break down in order to bring out the essential elements or structure. (To identify parts and relationships, and to interpret information to reach conclusions.)
Construct	Display information in a diagrammatic or logical form.
Create	To evolve from one's own thought or imagination, as a work or an invention.
Define	Give the precise meaning of a word, phrase, concept or physical quantity.
Demonstrate	Make clear by reasoning or evidence, illustrating with examples or practical application.
Describe	Give a detailed account or picture of a situation, event, pattern or process.
Design	Produce a plan, simulation or model.
Develop	To improve incrementally, elaborate or expand in detail. Evolve to a more advanced or effective state.
Evaluate	Make an appraisal by weighing up the strengths and limitations.
Explain	Give a detailed account including reasons or causes. (See also "Justify".)
Identify	Provide an answer from a number of possibilities. Recognize and state briefly a distinguishing fact or feature.
Justify	Give valid reasons or evidence to support an answer or conclusion. (See also "Explain".)
List	Give a sequence of brief answers with no explanation.
Outline	Give a brief account or summary.
Present	Offer for display, observation, examination or consideration.
Prioritize	Give relative importance to, or put in an order of preference.
State	Give a specific name, value or other brief answer without explanation or calculation.
Summarize	Abstract a general theme or major point(s).